How Much of the ROAD Masquerade Benchmark Is a Payload Shortcut?

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***Abstract*—Masquerade attacks, in which a compromised node silences a legitimate electronic control unit and transmits in its place at the correct rate, are regarded as the hardest attack class for in-vehicle Controller Area Network (CAN) intrusion detection because they leave message timing unchanged. The ROAD dataset supplies the community's reference masquerade captures. Those captures, however, are not recorded attacks: they are produced by deleting the victim's frames from the corresponding fabrication captures, which leaves the injected payloads untouched. We ask how much of the benchmark a rule with no learning can solve. Using four trivial detectors fitted only on attack-free traffic, we establish a payload-level difficulty floor for all 13 ROAD masquerade captures across five attack types. A per-identifier byte-range check reaches a window AUC-ROC of 1.00 on 7 captures (max speedometer, max coolant temperature, correlated signal) and detects 97–99% of attack windows at a 1% false-positive rate on the speedometer and correlated-signal captures. On these three attack types, published unsupervised masquerade detectors score at or below this floor even when their window parameters are tuned on the test captures (delta-over-trivial −0.26 to 0.00). Only the 6 reverse-light captures, which inject a legitimate state value at the wrong moment, resist every trivial rule (AUC 0.36–0.51). We trace the shortcut byte by byte to values that never occur in ambient traffic, and we recommend that CAN intrusion detection papers report per-attack performance relative to a trivial floor. Code and the floor table are released.**

***Index Terms*—Controller Area Network, intrusion detection, masquerade attack, benchmark evaluation, shortcut learning, automotive cyber security**

I. Introduction

The Controller Area Network (CAN) carries safety-relevant traffic in every modern consumer vehicle, yet it has no sender authentication: any node can transmit any identifier. Intrusion detection systems (IDS) for CAN are therefore an active research area, and they are compared on public datasets. Benchmarks such as Car-Hacking were later shown to be solvable from trivial timing features [3], [4], which moved the field toward ROAD [1], a dataset of real, physically verified attacks on a single vehicle.

Within ROAD, the *masquerade* captures are the reference hard case. A masquerade attacker suppresses the legitimate sender and injects forged frames at the original rate, so every timing-based detector fails by construction [3]. A line of signal-level detectors has been developed and benchmarked specifically on these captures [6], [7], [9]–[11].

The ROAD authors state, however, that the masquerade captures are *simulated*: they were produced from the fabrication captures by deleting the legitimate frames of the target identifier [1]. The injected frames themselves are unchanged, and the fabrication attacks inject extreme values (maximum speedometer, maximum coolant temperature). A rule as simple as "this byte never took this value in normal traffic" might therefore detect several masquerade captures. If so, scores on the benchmark measure less than the field assumes.

This paper measures that floor. We fit four trivial payload detectors on ROAD's ambient captures only and score all 13 masquerade captures. Our contributions are:

- The first payload-level trivial-baseline floor for the ROAD masquerade captures, connected to the frame-deletion procedure by which they were built (Sections III–V).
- Evidence that three of the five masquerade attack types are solved by a per-identifier byte-range check, and that published unsupervised masquerade detectors [6] score at or below this floor on them.
- A byte-level explanation of the shortcut and the identification of the reverse-light captures as the only part of the benchmark that resists trivial rules.
- A recommendation, with released code, to report delta-over-trivial per attack type.

II. Background and Related Work

Attacks. In a *fabrication* attack, injected frames are added beside the legitimate ones, roughly doubling the identifier's rate. In a *masquerade* attack, the legitimate sender is silenced and only forged frames remain, so the rate is preserved [1].

Trivial baselines and benchmark audits. Blevins et al. [3] benchmarked time-based IDS on ROAD and showed that simple timing statistics detect injection attacks well; masquerade attacks preserve timing and fall outside their scope. Heydari et al. [4] showed that frame-level classifiers on Car-Hacking and Survival exploit identifier shortcuts. Koltai et al. [5] found that single-dataset results do not predict cross-dataset performance. None of these establishes a payload-level floor for ROAD masquerade.

Masquerade detectors on ROAD. Moriano et al. [6] benchmark four unsupervised online detectors on the ROAD masquerade captures, namely matrix-correlation distribution (MCD) and correlation (MCC), Ganesan17 (G17) [8] and Moriano22 (M22) [7], under a sliding-window AUC-ROC protocol. Later work applies graph machine learning [9], state-space models [10] and per-identifier residualisation [11].

TABLE I
CLOSEST PRIOR WORK ON ROAD AND CAN BENCHMARK AUDITING.

Work	Data	Features	Trivial floor
Blevins et al. [3]	ROAD	timing	timing only
Heydari et al. [4]	Car-Hacking, Surv.	ID, payload	not ROAD
Koltai et al. [5]	7 CAN datasets	learned	no
Moriano et al. [6]	ROAD masquerade	signals	no
Marfo et al. [9]	ROAD masquerade	graph	no
Guerra et al. [12]	ROAD	ID, payload	no
This work	ROAD masquerade	payload	**yes**

TABLE II
BYTES WRITTEN BY EACH ATTACK (FROM ROAD METADATA) AND THE RANGE OF THAT BYTE IN AMBIENT TRAFFIC. * = VALUE NEVER SEEN IN AMBIENT.

Attack type	AID	Injected value [ambient range]
Max speedometer	0xD0	B5=255* [0,62]
Max coolant temp	0x4E7	B5=255* [79,162]
Correlated signal	0x6E0	B0=89* [0,53]; B2=69* [0,53]; B4=0 [0,53]; B6=255* [0,53]
Reverse light off	0xD0	B2=4 {4,12}
Reverse light on	0xD0	B2=12 {4,12}

Supervised comparisons [12] report near-perfect per-attack scores for an LSTM, and payload-based detectors such as CANdito [13] model decoded signals. All report absolute scores; none states what a trivial rule achieves on the same captures. Table I summarises the gap.

III. THE ROAD MASQUERADE CONSTRUCTION

ROAD [1], [2] contains 12 ambient captures (about 3 h) and 33 attack captures from one vehicle on a dynamometer. The 13 masquerade captures cover five attack types: max speedometer (3), max coolant temperature (1), correlated signal (3), reverse light off (3) and reverse light on (3). The authors describe their construction directly: "Using the fabrication captures we produce simulated masquerade attacks by removing the legitimate target ID frames preceding each injected frame" [1]. They also list this as a drawback, noting that no un-simulated masquerade data is publicly available.

We confirmed the construction in the data. For every masquerade capture, all non-target traffic is byte-identical to its fabrication twin, and the injected frames match in number up to two frames at the interval edges. Per-frame labels reconstructed from the capture metadata agree exactly with ROAD's own signal-file labels on all 13 captures.

Table II lists what each attack writes. In three types, the injected byte lies outside the range that byte ever takes in ambient traffic, highway driving included. In the correlated-signal attack, three of the four wheel-speed high bytes are out of range, so the signals are *not* individually plausible. In the reverse-light attacks, byte B2 of identifier 0xD0 only ever takes the values 4 and 12, and the attack writes one of them. The attack toggles the light "irrespective of the actual gear position" [6], which makes it a context anomaly: a legitimate value at the wrong moment.

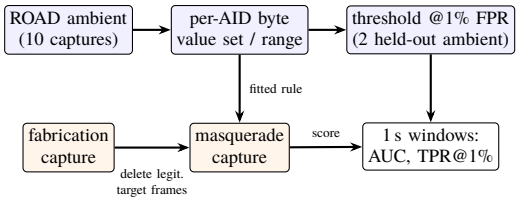


Fig. 1. Detectors are fitted on attack-free traffic only. ROAD's masquerade captures are fabrication captures with the legitimate target frames deleted, so the injected payloads are unchanged.

IV. METHOD

Trivial detectors. All four are fitted per arbitration identifier (AID) on ambient traffic only, and our code raises an error if an attack capture reaches the fitting path. T1 (value set) flags a frame if any byte takes a value never seen for that (AID, byte). T2 (range) scores the summed distance of each byte outside its ambient [min, max]. T3 (signal envelope) applies the same test to ROAD's decoded signals. T4 (payload novelty) flags a payload tuple never seen for that AID. Unseen AIDs receive the maximum score.

Protocol. We fit on 10 ambient captures and hold out two (*drive_basic_short*, *drive_extended_short*) to calibrate thresholds. Frame scores are averaged over 1 s windows with 0.25 s stride. As in [6], a window is positive if it overlaps the labelled injection interval, and we report window AUC-ROC. We also report the true-positive rate at the threshold that gives 1% false positives on the held-out ambient windows (TPR@1%), with 95% bootstrap confidence intervals over windows (10,000 resamples). Figure 1 summarises the pipeline.

Controls and comparators. As a learned control we fit one isolation forest per AID on the eight payload bytes (200 trees, three seeds). The code of [6] depends on a module that is not in its public repository, so we *quote* its Table 2: per attack type, the mean AUC over its window grid and the best grid cell. The best cell is selected on the test captures and is therefore optimistic. We define *delta-over-trivial* as the best published AUC minus the T2 AUC, averaged over the captures of each type.

V. RESULTS

The floor. Table III gives window AUC-ROC per capture. On all 7 speedometer, coolant and correlated-signal captures, the simplest rule (T2) reaches 1.00, as do T1 and T3 with one exception (T1 0.98 on coolant). At the operating point calibrated on held-out ambient, T2 detects 97–99% of attack windows on the speedometer and correlated-signal captures. The coolant capture ranks perfectly yet is missed at 1% FPR: it contains only 43 injected frames, and its excursion (255 against an ambient maximum of 162) is smaller than benign excursions on other identifiers in the held-out captures. On the extreme-value captures, every fabrication twin scores the same as its masquerade capture (largest difference 0.00), as the construction predicts. On the 6 reverse-light captures, no trivial rule rises meaningfully above chance (T2 0.36–0.51). Figure 2

TABLE III

WINDOW AUC-ROC OF TRIVIAL DETECTORS (FITTED ON AMBIENT ONLY) AND THE ISOLATION-FOREST CONTROL (IF, MEAN OF 3 SEEDS) PER MASQUERADE CAPTURE; TPR@1%: % OF ATTACK WINDOWS DETECTED BY T2 AT 1% FPR ON HELD-OUT AMBIENT. BOLD: ≥ 0.95.

Capture	T1	T2	T3	T4	IF	TPR@1%
speedo-1	**1.00**	**1.00**	**1.00**	0.67	0.53	98
speedo-2	**1.00**	**1.00**	**1.00**	0.92	0.78	98
speedo-3	**1.00**	**1.00**	**1.00**	0.68	0.91	99
coolant	**0.98**	**1.00**	**1.00**	0.59	0.80	0
corr-1	**1.00**	**1.00**	**1.00**	0.68	0.87	97
corr-2	**1.00**	**1.00**	**1.00**	0.04	0.33	99
corr-3	**1.00**	**1.00**	**1.00**	0.25	0.99	98
rl-off-1	0.40	0.50	0.50	0.46	0.38	0
rl-off-2	0.79	0.50	0.72	0.71	0.39	0
rl-off-3	0.49	0.45	0.50	0.42	0.34	0
rl-on-1	0.37	0.36	0.52	0.41	0.83	0
rl-on-2	0.41	0.45	0.40	0.89	0.88	0
rl-on-3	0.45	0.51	0.53	0.83	0.86	0

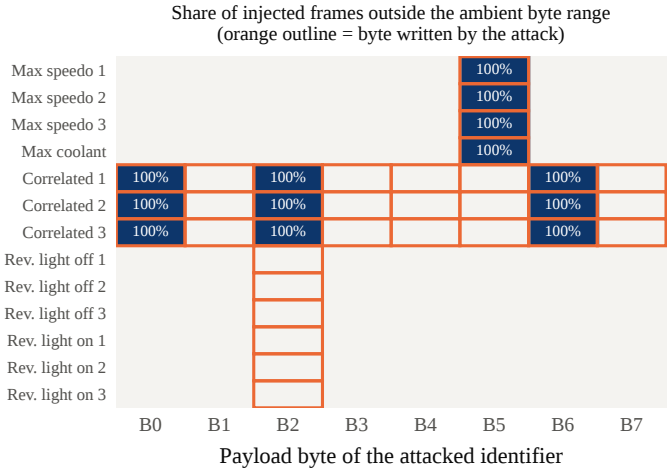


Fig. 2. Share of injected frames whose byte lies outside the ambient range of the attacked identifier. The shortcut sits exactly on the bytes written by the attack; reverse-light frames never leave the range.

shows why: every injected frame of the three extreme types violates the ambient range on exactly the bytes the metadata says were written, and no reverse-light frame does.

Figure 3 shows the same effect over time: the share of flagged frames steps up exactly over the injection interval for the speedometer and correlated-signal attacks and does not move for reverse light.

Against published detectors. Table IV and Fig. 4 compare the floor with [6]. On max speedometer, max coolant and correlated signal, the best published detector, even with test-tuned window parameters, ties or trails the range check (delta -0.26 to 0.00). On average over its window grid it trails by much more. The published methods earn their keep only on the reverse-light types, where they lead the trivial floor by +0.25 to +0.28.

A learned payload model is not a free improvement. The per-AID isolation forest is weaker than the range rule on

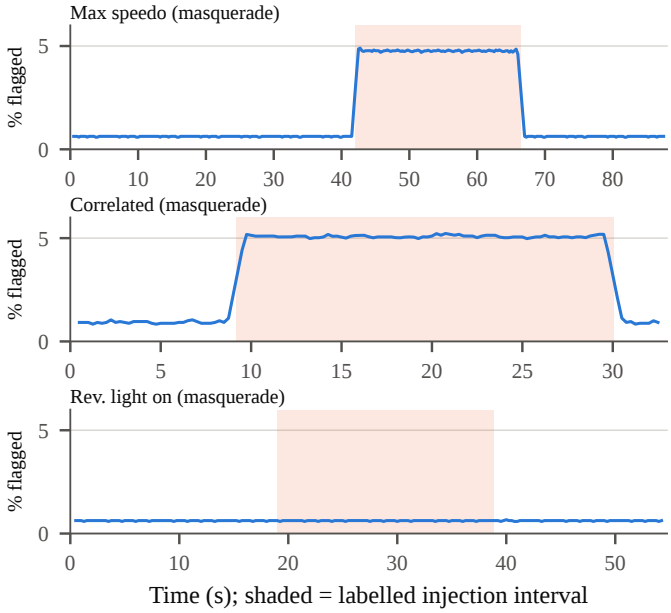


Fig. 3. Share of frames flagged by T2 per 1 s window for one capture of three attack types.

TABLE IV

TRIVIAL FLOOR (T2, MEAN OVER CAPTURES) VERSUS THE BEST OF FOUR PUBLISHED UNSUPERVISED DETECTORS [6] (QUOTED). DELTA = TUNED MAX − T2.

Attack type	T2	Published mean	Published max	Delta
Max speedometer	1.00	0.64 (G17)	0.74 (G17)	−0.26
Max coolant temp	1.00	0.66 (M22)	1.00 (M22)	0.00
Correlated signal	1.00	0.71 (MCC)	0.88 (MCC)	−0.12
Reverse light off	0.48	0.68 (MCD)	0.76 (MCD)	+0.28
Reverse light on	0.44	0.59 (M22)	0.69 (M22)	+0.25

the extreme-value types (e.g. 0.74 on speedometer), because one out-of-range byte barely moves a density score over eight bytes. It is the only payload model that separates reverse-light-on (0.86): the value 12 is rare in ambient traffic and rarely co-occurs with forward-driving bytes. On reverse-light-off it falls below chance (0.37), because the injected value 4 is the common state.

VI. ABLATION AND SENSITIVITY

Where the signal lives. Restricting T2 to a single byte reproduces the full result on exactly the written bytes (B5 for speedometer and coolant; B0, B2, B6 for correlated signal) and yields chance elsewhere. Ignoring the attacked identifier drops the extreme-value AUCs from 1.00 to 0.24–0.71, so physical side-effects in other identifiers carry far less signal than the payload itself. Normalising excursions by range width leaves every AUC unchanged.

Robustness. Figure 5 varies the window length (0.25–2 s) and the share of each ambient capture used for fitting (10–100%). The extreme-value AUCs never fall below 1.00, so the floor holds with a tenth of the ambient data. Relaxing T3 to

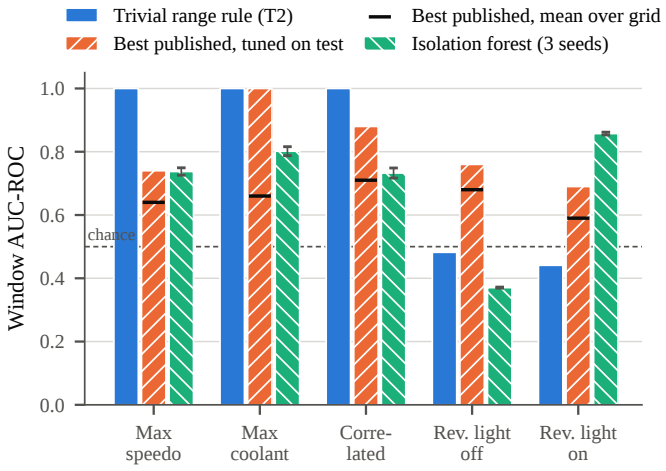


Fig. 4. Trivial range rule versus the best published detector per attack type [6] and the isolation-forest control. Dashes mark the published mean over the window grid.

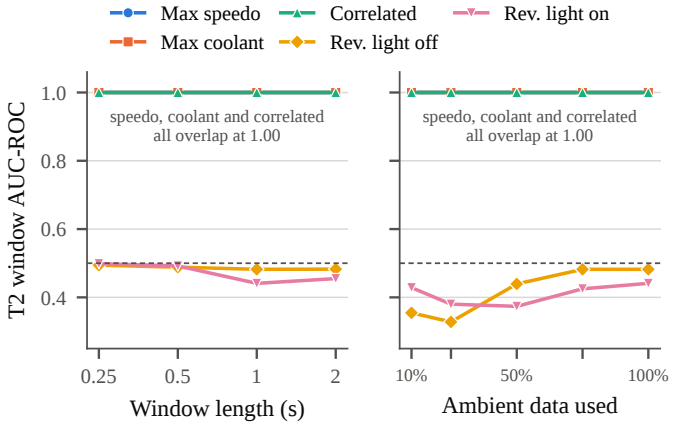


Fig. 5. T2 window AUC-ROC per attack type (mean over captures) versus window length and amount of ambient data used for fitting.

the [p1, p99] envelope also leaves them at 0.97 or above. The reverse-light AUCs stay at or below chance throughout.

VII. Discussion and Limitations

What this shows, and what it does not. We do not claim that masquerade detection is easy in real vehicles. A real attacker controls the payload and would avoid out-of-range values. Our claim concerns the benchmark: the speedometer, coolant and correlated-signal captures inherit extreme payloads from their fabrication originals, so they measure out-of-range detection rather than masquerade detection. High scores on them, including near-perfect supervised scores [12], are consistent with a shortcut.

Recommendations. (i) Report masquerade results per attack type, not pooled. (ii) Report delta-over-trivial against the T2 floor, which takes a few lines of code. (iii) Treat the six reverse-light captures as the informative subset for masquerade detection on ROAD. (iv) New masquerade datasets should inject in-range values, as the ROAD authors themselves call for un-simulated masquerade data [1].

Limitations. ROAD covers one vehicle on a dynamometer. Overlapping windows make the bootstrap intervals optimistic. Comparator numbers are quoted rather than re-run; they average over the captures of each type and use sample-count windows on 100 Hz signals, whereas ours are time-based, so the deltas are indicative. The test-tuned published maxima favour the comparators, which strengthens rather than weakens our conclusion. A second corpus with independently simulated masquerades [14] would test whether the shortcut is specific to ROAD; we leave this to future work.

VIII. Conclusion

Seven of ROAD's 13 masquerade captures are solved by checking whether a byte left its ambient range, and on those captures published unsupervised detectors do not beat that rule. The effect follows from how the captures were built. The reverse-light captures, which inject a plausible value at the wrong moment, are the part of the benchmark that still measures masquerade detection. We release our floor so that future work can report against it.

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